

Paso 1

```

melissa@melissa-Inspiron-3505:~$ sudo apt update && sudo apt install easy-rsa -y
Des:1 https://dl.google.com/linux/chrome/deb stable InRelease [2.548 B]
Des:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Obj:3 http://archive.ubuntu.com/ubuntu noble InRelease
Des:4 https://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1.410 B]
Obj:5 http://linux.dropbox.com/ubuntu noble InRelease
Des:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Des:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1.009 kB]
Des:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Des:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [213 kB]
Des:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1.263 kB]
Des:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46,4 kB]
Des:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1.449 kB]
Des:13 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [336 kB]
Des:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1.207 kB]
Des:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [242 kB]
Des:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76,3 kB]
Des:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [180 kB]
Des:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1.687 kB]
Des:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [338 kB]
Des:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Des:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Des:22 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5.776 B]
Des:23 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12,6 kB]
Descargados 8.837 kB en 4s (2.479 kB/s)
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se pueden actualizar 159 paquetes. Ejecute «apt list --upgradable» para verlos.
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.

```

```
melissa@melissa-Inspiron-3505:~$ ls /usr/share/easy-rsa/
easysra  openssl-easysra.cnf  vars.example  x509-types
```

Paso 2

```
melissa@melissa-Inspiron-3505:~$ mkdir ~/pki-ci0123 && cd ~/pki-ci0123
melissa@melissa-Inspiron-3505:~/pki-ci0123$ /usr/share/easy-rsa/easyrsa init-pki

Notice
-----
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /home/melissa/pki-ci0123/pki

Using Easy-RSA configuration:
* undefined
```

Paso 3

[illegible]

You are about to be asked to enter information that will be incorporated into your certificate request.

What you are about to enter is what is called a Distinguished Name or a DN. There are quite a few fields but you can leave some blank

For some fields there will be a default value,

If you enter '.', the field will be left blank.

```
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:CI0123 CA
```

Notice

CA creation complete. Your new CA certificate is at:

```
* /home/melissa/pki-ci0123/pki/ca.crt
```

Paso 4

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ /usr/share/easy-rsa/easyrsa gen-dh
No Easy-RSA 'vars' configuration file exists!

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
Generating DH parameters, 2048 bit long safe prime
...+.....
.....
.....
.....
.....+.....+.....
.....+.....
.....+.....
```

DH parameters appear to be ok.

Notice

```
DH parameters of size 2048 created at:
* /home/melissa/pki-ci0123/pki/dh.pem
```

Paso 5

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ /usr/share/easy-rsa/easyrsa build-server-full redes nopass
No Easy-RSA 'vars' configuration file exists!
```

Using SSL:

```
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
```

[illegible]

Notice

```
Private-Key and Public-Certificate-Request files created.
```

Your files are:

```
* req: /home/melissa/pki-ci0123/pki/reqs/redes.req
```

```
* key: /home/melissa/pki-ci0123/pki/private/reds.key
```

You are about to sign the following certificate:

Request subject, to be signed as a server certificate

```
for '825' days:
```

subject=

```
commonName      = redes
```

```
Type the word 'yes' to continue, or any other input to abort.
```

Confirm request details: yes

```
Using configuration from /home/melissa/pki-ci0123/pki/openssl-easyrsa.cnf
```

```
Enter pass phrase for /home/melissa/pki-ci0123/pki/private/ca.key:
```

```
Check that the request matches the signature
```

Signature ok

The Subject's Distinguished Name is as follows

```
commonName      :ASN.1 12:'redes'
```

Certificate is to be certified until Dec 18 23:04:17 2028 GMT (825 days)

```
The Subject's Distinguished Name is as follows
commonName             :ASN.1 12:'redes'
Certificate is to be certified until Dec 18 23:04:17 2028 GMT (825 days)
```

```
Write out database with 1 new entries
Database updated
```

```
Notice
```

```
-----
```

```
Certificate created at:
```

```
* /home/melissa/pki-ci0123/pki/issued/redes.crt
```

```
Notice
```

```
-----
```

```
Inline file created:
```

```
* /home/melissa/pki-ci0123/pki/inline/redes.inline
```

Paso 6

```
Private-Key and Public-Certificate-Request files created.
```

```
Your files are:
```

```
* req: /home/melissa/pki-ci0123/pki/reqs/cliente01.req
```

```
* key: /home/melissa/pki-ci0123/pki/private/cliente01.key
```

```
You are about to sign the following certificate:
```

```
Request subject, to be signed as a client certificate
```

```
for '825' days:
```

```
subject=
```

```
commonName             = cliente01
```

```
Type the word 'yes' to continue, or any other input to abort.
```

```
Confirm request details: yes
```

```
Using configuration from /home/melissa/pki-ci0123/pki/openssl-easyrsa.cnf
```

```
Enter pass phrase for /home/melissa/pki-ci0123/pki/private/ca.key:
```

```
Check that the request matches the signature
```

```
Signature ok
```

```
The Subject's Distinguished Name is as follows
```

```
commonName             :ASN.1 12:'cliente01'
```

```
Certificate is to be certified until Dec 18 23:07:02 2028 GMT (825 days)
```

```
Write out database with 1 new entries
```

```
Database updated
```

```
Notice
```

```
-----
```

```
Certificate created at:
```

```
* /home/melissa/pki-ci0123/pki/issued/cliente01.crt
```

```
Notice
```

```
-----
```

```
Inline file created:
```

```
* /home/melissa/pki-ci0123/pki/inline/cliente01.inline
```

```

melissa@melissa-Inspiron-3505:~/pki-ci0123$ ls -la pki/private pki/issued
pki/issued:
total 24
drwx----- 2 melissa melissa 4096 sep 15 17:07 .
drwx----- 8 melissa melissa 4096 sep 15 17:07 ..
-rw----- 1 melissa melissa 4492 sep 15 17:07 cliente01.crt
-rw----- 1 melissa melissa 4595 sep 15 17:04 redes.crt

pki/private:
total 20
drwx----- 2 melissa melissa 4096 sep 15 17:06 .
drwx----- 8 melissa melissa 4096 sep 15 17:07 ..
-rw----- 1 melissa melissa 1854 sep 15 16:55 ca.key
-rw----- 1 melissa melissa 1704 sep 15 17:06 cliente01.key
-rw----- 1 melissa melissa 1700 sep 15 17:03 redes.key

```

```

melissa@melissa-Inspiron-3505:~/pki-ci0123$ openssl x509 -in pki/ca.crt -text -noout | head -20
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            78:2d:54:0b:5b:1b:22:e6:0d:18:36:82:68:97:e2:3a:02:5f:b7:3e
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN = CI0123 CA
        Validity
            Not Before: Sep 15 22:56:00 2026 GMT
            Not After : Sep 12 22:56:00 2036 GMT
        Subject: CN = CI0123 CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            Public-Key: (2048 bit)
            Modulus:
                00:d6:f5:3b:e8:47:e8:75:8d:75:59:97:8b:90:b4:
                75:58:8f:01:0d:d8:f6:f5:a9:b8:b5:00:99:15:ac:
                91:7e:da:2b:8e:d2:8f:14:be:9e:65:c7:f6:9c:f0:
                3a:41:28:7f:1c:88:50:6c:f1:97:2e:1b:57:f7:ff:
                89:50:eb:4b:02:5f:3b:12:0c:40:d1:d0:eb:a8:25:

```

Construir un certificado auto-firmado para Apache:

Paso 1

```

melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo apt install apache2 -y
[sudo] contraseña para melissa:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
  libgl1-amd64 libglapi-mesa libllvm17t64 python3-netifaces
Utilice «sudo apt autoremove» para eliminarlos.
Se instalarán los siguientes paquetes adicionales:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
Paquetes sugeridos:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
Se instalarán los siguientes paquetes NUEVOS:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
0 actualizados, 8 nuevos se instalarán, 0 para eliminar y 159 no actualizados.
Se necesita descargar 1.912 kB de archivos.
Se utilizarán 7.490 kB de espacio de disco adicional después de esta operación.
Des:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Des:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7.1 [92,7 kB]
Des:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7.1 [11,2 kB]
Des:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7.1 [9,120 B]
Des:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.15 [1.338 kB]
Des:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.15 [163 kB]
Des:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.15 [99,7 kB]
Des:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.15 [90,2 kB]
Descargados 1.912 kB en 2s (843 kB/s)
Seleccionando el paquete libapr1t64:amd64 previamente no seleccionado.
(Leyendo la base de datos ... 22017 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../8-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Desempaquetando libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Seleccionando el paquete libaprutil1t64:amd64 previamente no seleccionado.
Preparando para desempaquetar .../1-libaprutil1t64_1.6.3-1.1ubuntu7.1_amd64.deb ...
Desempaquetando libaprutil1t64:amd64 (1.6.3-1.1ubuntu7.1) ...
Seleccionando el paquete libaprutil1-dbd-sqlite3:amd64 previamente no seleccionado.
Preparando para desempaquetar .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7.1_amd64.deb ...
Desempaquetando libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7.1) ...
Seleccionando el paquete libaprutil1-ldap:amd64 previamente no seleccionado.

```

Paso 2

```
melissa@melissa-Inspiron-3505: ~$ sudo systemctl status apache2
[sudo] contraseña para melissa:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-15 17:14:20 CST; 56min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 127110 (apache2)
     Tasks: 55 (limit: 5561)
  Memory: 5.6M (peak: 6.9M)
     CPU: 283ms
   CGroup: /system.slice/apache2.service
           └─127110 /usr/sbin/apache2 -k start
             └─127112 /usr/sbin/apache2 -k start
               └─127113 /usr/sbin/apache2 -k start

sep 15 17:14:20 melissa-Inspiron-3505 systemd[1]: Starting apache2.service - The Apache HTTP Server...
sep 15 17:14:20 melissa-Inspiron-3505 apachectl[127109]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using
sep 15 17:14:20 melissa-Inspiron-3505 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

Paso 3

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
    systemctl restart apache2
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo systemctl restart apache2
melissa@melissa-Inspiron-3505:~/pki-ci0123$
```

Paso 4

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ /usr/share/easy-rsa/easyrsa build-server-full apache nopass
No Easy-RSA 'vars' configuration file exists!
```

```
Using SSL:  
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)  
.....  
+.....+
```

```
Using configuration from /home/melissa/pki-ci0123/pki/openssl-easyrsa.cnf
Enter pass phrase for /home/melissa/pki-ci0123/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName             :ASN.1 12:'apache'
Certificate is to be certified until Dec 19 00:17:17 2028 GMT (825 days)
```

```
Write out database with 1 new entries
Database updated
```

Notice

```
Certificate created at:
* /home/melissa/pki-ci0123/pki/issued/apache.crt
```

Notice

```
Inline file created:
* /home/melissa/pki-ci0123/pki/inline/apache.inline
```

Paso 5

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo mkdir -p /etc/apache2/ssl
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo cp pki/issued/apache.crt /etc/apache2/ssl/
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo cp pki/private/apache.key /etc/apache2/ssl/
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo chmod 600 /etc/apache2/ssl/apache.key
melissa@melissa-Inspiron-3505:~/pki-ci0123$ ls -la /etc/apache2/ssl/
total 20
drwxr-xr-x 2 root root 4096 sep 15 18:19 .
drwxr-xr-x 9 root root 4096 sep 15 18:19 ..
-rw----- 1 root root 4601 sep 15 18:19 apache.crt
-rw----- 1 root root 1704 sep 15 18:19 apache.key
```

Paso 6

```
# A self-signed (snakeoil) certificate can be created by installing
# the ssl-cert package. See
# /usr/share/doc/apache2/README.Debian.gz for more info.
# If both key and certificate are stored in the same file, only the
# SSLCertificateFile directive is needed.
SSLCertificateFile      /etc/apache2/ssl/apache.crt
SSLCertificateKeyFile   /etc/apache2/ssl/apache.key
```

Paso 7

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo a2ensite default-ssl
Enabling site default-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo systemctl restart apache2
melissa@melissa-Inspiron-3505:~/pki-ci0123$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-15 18:26:58 CST; 9s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 129507 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 129512 (apache2)
       Tasks: 55 (limit: 5561)
      Memory: 7.0M (peak: 7.8M)
         CPU: 66ms
    CGroup: /system.slice/apache2.service
            └─129512 /usr/sbin/apache2 -k start
              129514 /usr/sbin/apache2 -k start
              129515 /usr/sbin/apache2 -k start

sep 15 18:26:58 melissa-Inspiron-3505 systemd[1]: Starting apache2.service - The Apache HTTP Server...
sep 15 18:26:58 melissa-Inspiron-3505 apachectl[129511]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using
sep 15 18:26:58 melissa-Inspiron-3505 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

Paso 8

```
melissa@melissa-Inspiron-3505:~/pki-ci0123$ curl -vk https://localhost/ 2>&1 | grep -A5 "Server certificate"
* Server certificate:
* subject: CN=apache
* start date: Sep 16 00:17:17 2026 GMT
* expire date: Dec 19 00:17:17 2028 GMT
* issuer: CN=CI0123 CA
* SSL certificate verify result: unable to get local issuer certificate (20), continuing anyway.
```

Prueba desde el navegador con <https://localhost/>

Certificado	
apache	
Nombre del asunto	
Nombre común	apache
Nombre del emisor	
Nombre común	CI0123 CA
Validez	
No antes	Wed, 16 Sep 2026 00:17:17 GMT
No después	Tue, 19 Dec 2028 00:17:17 GMT
Nombres alternativos del sujeto	
Nombre de la DNS	apache
Información de clave pública	